Concept Note

Front-Consistency Vetoing of Impulsive RFI in Radio Air-Shower Arrays: A Spatial-Signature Trigger Layer, Pre-Specified

Nasir Khan^1

^1 Independent Researcher — Foundations of Physics & Cosmology,
Wana, South Waziristan, Khyber Pakhtunkhwa, Pakistan
Email: Nasirk684@gmail.com · ORCID: 0009-0002-6006-0796

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1. Problem

Radio detectors of extensive air showers — AERA at the Pierre Auger Observatory, LOFAR, and the planned BEACON and GRAND arrays — trigger on nanosecond broadband pulses in a busy RF environment. RFI rejection today rests on two legs: notch filters for the narrowband crowd, and direction-of-arrival cuts that run later, in offline reconstruction. Both work, and both leave something on the table. Impulsive broadband RFI slips past the notches and still fires the

trigger, spending data volume and reconstruction time on events that were never showers. And the real-time side of the problem — a keep/reject decision made at the trigger, under power constraints, in a falsifiable form — remains open, particularly for the autonomous arrays where nobody is watching.

2. Proposal: front-consistency veto

A shower lights up the array as a coherent front. The delays between antennas carry the geometry of a nearly planar, slightly curved wavefront arriving from the reconstructed shower direction. A fixed RFI transmitter writes a different signature: the delays of its direction, always the same one.

Both fire many antennas within a short window, so counting coincidences cannot tell them apart. This is worth stating plainly, because it inverts the logic of the companion swarm-veto protocol, where genuine events were node-local and interference was everywhere at once. In a radio array the signal is the thing that arrives everywhere, and the interference is the thing with a direction. What separates them is not whether the antennas fire together but the pattern of their delays.

The proposed layer therefore accepts a multi-antenna event only when its delay vector matches, at a pre-specified confidence, a front hypothesis reconstructed from the same data — and rejects it otherwise. Point-source RFI fails this test by construction. The exceptions are bounded and declared: a

transmitter that happens to sit along the reconstructed shower direction mimics a front and survives. That blind region is the direct analog of the equidistant-geometry blind region in the companion protocol, and it is treated the same way — quantified in advance, not discovered in discussion.

3. Analogy mapping

Table 1 carries the whole argument in one view: every element of the locked swarm protocol has a radio-array counterpart.

Swarm veto protocol	Radio air-shower array
Detector node (ESP32 + scintillator/SiPM)	Antenna / station channel
Swarm-global EMI, simultaneous at all nodes	Point-source RFI, coherent across the array with

one direction's delays
Node-local genuine event Planar-front shower pulse, delays of the reconstructed shower
Coincidence window Δt Inter-antenna delay window from the front model and array geometry
Voting rule (≥2 nodes → veto) Front-consistency test of the delay vector against H_front
Equidistant-source blind region Transmitter directions degenerate with the reconstructed shower
Per-node Poisson preservation Trigger-rate preservation; no azimuth or energy bias
Falsifiers S1–S5 Falsifiers F1–F5 (Table 2)
Seed-684 computational validation CoREAS showers plus injected RFI models

4. Falsifier framework

The criteria below are proposed defaults. The numbers get fixed in the planned pre-registration, before any simulation data exist — the same discipline the companion protocol family has kept throughout.

Table 2. Proposed falsifiers.

ID Claim protected Proposed criterion (fixed at pre-registration)
F1 Impulsive RFI rejection ≥ 95% of injected point-source RFI rejected at trigger level, per RFI class (carrier-like, periodic, random-impulsive)
F2 Signal efficiency ≥ 90% of simulated showers retained that a pure energy trigger keeps, zenith 0–60°, uniform azimuth
F3 Selection bias Retained-sample azimuth and energy distributions match input at pre-specified confidence; no directional efficiency structure beyond the declared blind region

F4 Threshold portability Front-consistency confidence and delay-window parameters set on a calibration subset, applied unchanged to all test data
F5 Degraded-array robustness F1–F3 conclusions hold with one and two antennas randomly excluded

5. Validation plan

The first step is a computational study in the exact pattern of the companion validation: CoREAS-simulated shower pulses, parameterized RFI, an array model, and a seed-pooled notebook that regenerates every number on a Run-All. F1–F5 are measured and reported as an engineering analogue — control-only, in the language this program has used since the beginning. If and only if those

falsifiers pass, the second step is to walk up to an operating collaboration — the AERA analysis group, BEACON, GRANDproto — with the locked protocol and the computational result in hand, and propose application to public data or a deliberately RFI-rich run. No deployment claim is made here, and the plan does not make one later without hardware behind it.

6. Relation to existing methods

This is a complement to notch filtering and interferometric DoA reconstruction, not a replacement. Those methods work; the proposal simply moves the same delay information up to the trigger decision, where it runs in real time, and wraps it in a falsifier framework that trigger-level RFI rejection in the

literature generally does not carry.
Where an array's offline pipeline already removes a class of RFI, F1 is scored against what survives to the trigger — the residual that actually costs the array its time and bandwidth.

7. Status and lineage

This note fixes the mechanism and the falsifier framework ahead of any data, as the program's protocol family has done at every step: QCID-Sigma v5.4, the muon pre-registration, the propulsion-EMI protocol, the swarm veto protocol and its computational validation. The planned pre-registration and simulation study will cite this note by version.

Declarations

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conceived the mechanism and wrote the note.
Correspondence: Nasirk684@gmail.com

References (indicative)

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